

CYBER SECURITY COURSE

Linux Memory and Process Forensics

Professional Case Report 07

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Report date	September 24, 2026
Case ID	CSC-2026-07
Discipline	Digital forensics
Overall risk	High
Evidence status	Laboratory methodology documented

Purpose

Recover volatile evidence and attribute suspicious processes without relying only on files visible in the normal directory tree.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

A Linux host contained hidden processes, deleted executables, volatile shared-memory artifacts and suspicious sockets. The host must be treated as a live evidence source. Process state, file descriptors and tmpfs contained information that ordinary disk review could miss.

Case Profile

Scenario	A Linux host contained hidden processes, deleted executables, volatile shared-memory artifacts and suspicious sockets.
Environment	Linux forensic simulation with procfs, tmpfs, raw image and socket artifacts.
Tools	ps, lsof, procfs, strings, md5sum, ss, blkid, stat

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Enumerated processes and inspected environment variables through /proc.
2. Used lsof +L1 to locate unlinked binaries still held open by a process.
3. Recovered deleted content through live file descriptors.
4. Inspected /dev/shm and raw images, then calculated evidence hashes.
5. Mapped listening sockets to owning processes.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Unlinked executable	High	A running process can retain a deleted binary through an open file descriptor.
F02	Volatile shared memory	High	Artifacts in /dev/shm may disappear after reboot and require early collection.
F03	Network listener attribution	High	An unusual listener must be tied to its PID and executable before containment.
F04	Integrity verification	Medium	Hashes establish that collected images and recovered files remain unchanged.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
ps aux | grep [s]ys_worker  
sudo lsof +L1
```

```
sudo find /proc/*/fd -ls | grep '(deleted)'  
sudo ss -ltp  
md5sum /dev/shm/hidden_disk.img
```

Risk Assessment

Risk	Level	Operational impact
Volatile evidence loss	High	Reboot destroys memory-backed artifacts
Hidden process execution	High	Deleted binaries can continue operating without a normal path

Recommendations

1. Collect process, socket and /proc evidence before shutdown.
2. Monitor execution from tmpfs locations.
3. Alert on deleted executables with active file descriptors.
4. Centralize logs so the local attacker cannot remove all evidence.

Knowledge Demonstrated

- Linux live response
- procfs analysis
- Deleted-file recovery
- Socket attribution

Conclusion

The host must be treated as a live evidence source. Process state, file descriptors and tmpfs contained information that ordinary disk review could miss.

Hands On Lab Practical 07

Completed as part of the authorized cyber security course practical work.